

# Strategic Product Placement Analysis Project Report

## Abstract

Strategic Product Placement Analysis is a Business Intelligence project that analyzes retail product placement and its impact on sales using Tableau. A retail dataset from Kaggle was imported into Tableau Desktop for preprocessing and visualization. Interactive dashboards and Tableau Stories were created to understand pricing, promotions, competitor pricing, product positioning, seasonal demand, customer demographics and sales performance. The completed dashboard and story were published to Tableau Public and embedded into a responsive Bootstrap website developed using HTML5, CSS3 and Visual Studio Code. This project demonstrates how Business Intelligence tools transform raw retail data into meaningful business insights.

## Introduction

Business Intelligence helps organizations make informed decisions by converting raw data into useful information. In this project Tableau Desktop was used to analyze a retail dataset and understand how pricing, promotions, competitor pricing, product position and seasonal demand influence sales. The project covers the complete BI workflow including data collection, preprocessing, visualization, dashboard creation, story development, publishing and website integration.

## Problem Statement

Retail stores must determine the best shelf positions and pricing strategies to maximize sales. Product visibility, promotional offers, competitor pricing and customer preferences all influence purchasing decisions. Manual analysis is difficult because of the large amount of retail data. This project uses Tableau to create dashboards and stories that help retailers make data-driven decisions.

## Objectives

The objectives are to collect a retail dataset from Kaggle, connect it with Tableau Desktop, preprocess and validate the data, create meaningful visualizations, build an interactive dashboard, create a Tableau Story, publish the workbook to Tableau Public and integrate it into a Bootstrap website using HTML5, CSS3 and Visual Studio Code.

## Scope of the Project

The project focuses on analyzing historical retail data to identify sales trends, pricing strategies, promotions and customer behavior. Interactive dashboards help managers explore data through filters and improve product placement decisions. The same workflow can be extended to real-time business environments.

## Dataset Collection

The dataset was downloaded from Kaggle in CSV format. It contains approximately 1000 product records with attributes including Product Category, Product Price, Competitor Price, Product Position, Promotion, Seasonal Demand, Customer Demographics, Foot Traffic and Sales Volume. The CSV file was imported into Tableau Desktop using the Text File connector.

## Dataset Description

The dataset contains structured numerical and categorical fields suitable for Business Intelligence analysis. Each row represents one product record while columns describe product attributes. The dataset provides sufficient information to analyze pricing, promotions, customer behavior and sales trends.

## **Software, Tools and Technologies Used**

Kaggle was used for dataset collection. Tableau Desktop was used for preprocessing, visualization, dashboards and stories. Tableau Public was used for publishing. Visual Studio Code, HTML5, CSS3 and Bootstrap 5 were used to develop the website. Google Chrome was used for testing and Google Drive for sharing the demonstration video.

## **Connecting the Dataset to Tableau Desktop**

The retail dataset downloaded from Kaggle was imported into Tableau Desktop using the Text File connector. After selecting the CSV file, Tableau displayed the Data Source page where all fields and records were loaded automatically. Around 1000 records were successfully imported. This step ensured that the dataset was ready for preprocessing and visualization. Tableau's Data Source view also allowed verification of columns, field names, and record counts before analysis.

## **Data Review and Exploration**

After importing the dataset, each field was reviewed to understand its structure and business meaning. Tableau automatically categorized fields into Dimensions and Measures based on their data types. Product Category, Promotion Status, Product Position, and Customer Demographics were identified as dimensions, while Product Price, Competitor Price, Sales Volume, and Foot Traffic were measures. This review helped determine which charts and analyses would be most suitable for the project.

## **Data Cleaning and Preparation**

Although the dataset was already clean, basic preprocessing was performed. Missing values and duplicate records were checked to ensure data quality. Numeric fields were verified for consistency, and unnecessary formatting issues were removed. Since the dataset was well structured, no major cleaning operations were required. These preparation steps ensured reliable results during visualization.

## **Field Renaming and Formatting**

Field names were reviewed to improve readability inside Tableau. Numerical fields such as Product Price, Competitor Price, and Price Difference were formatted using appropriate number formats. Tableau's Default Properties option was used to display values consistently. Proper formatting improves dashboard presentation and makes visualizations easier to interpret.

## **Data Validation**

Validation was performed to ensure the imported data matched the source dataset. The Number of Records field confirmed that approximately 1000 records were available. Summary statistics such as average Product Price and Competitor Price were checked. Calculated fields were tested to verify that they produced expected outputs. This process increased confidence in the accuracy of the dashboard.

## **Calculated Fields**

Four calculated fields were created in Tableau: Price Difference, Promotion Status, Discount Category, and Sales per Price. Price Difference compared product prices with competitor prices. Promotion Status converted raw promotion values into user-friendly labels. Discount Category grouped products based on price differences, while Sales per Price measured sales efficiency. These calculated fields enhanced the analytical capabilities of the dashboard.

## **Filters Used**

Interactive filters were added to improve dashboard usability. Product Category, Promotion Status, Product Position, Customer Demographics, Seasonal Demand, and Foot Traffic filters allow users to focus on specific subsets of data. Filters automatically updated all visualizations in the dashboard, making analysis flexible and interactive.

## **Importance of Data Preparation**

Proper data preparation forms the foundation of Business Intelligence projects. Reviewing, validating, formatting, and organizing the dataset ensures that dashboards present accurate and meaningful information. These preprocessing activities reduced the possibility of errors and improved the reliability of business insights generated through Tableau

### **Visualization 1 – Sales Volume by Product Category (Bar Chart)**

The first visualization created was a Bar Chart comparing Sales Volume across Product Categories. Product Category was placed on the Columns shelf while Sales Volume was placed on the Rows shelf. The chart clearly highlighted which categories generated the highest and lowest sales. Appropriate labels, colors, and tooltips were added to improve readability. This visualization helps retailers identify high-performing categories and supports better inventory planning and product placement decisions.

### **Visualization 2 – Promotion Analysis (Stacked Bar Chart)**

A Stacked Bar Chart was developed to compare promoted and non-promoted products. Product Category and Sales Volume were used along with Promotion Status as the color field. This visualization showed the impact of promotional campaigns on product sales. Retail managers can use these insights to evaluate marketing effectiveness and optimize promotional strategies.

### **Visualization 3 – Product Position Distribution (Donut Chart)**

A Donut Chart was used to display the distribution of products based on their placement within the store. Product Position values such as Front Shelf, Middle Shelf, and Back Shelf were represented as segments. Percentage labels and colors improved understanding. This visualization helps retailers understand shelf allocation and improve product visibility.

### **Visualization 4 – Competitor Price Comparison (Scatter Plot)**

A Scatter Plot compared Product Price with Competitor Price. Each point represented an individual product, making it easy to identify products priced above or below competitors. This visualization supports pricing strategy decisions and helps retailers remain competitive in the market.

### **Visualization 5 – Seasonal Demand Analysis (Line Chart)**

A Line Chart was created to analyze Sales Volume across different Seasonal Demand categories. The visualization clearly displayed fluctuations in demand over different seasons. Retailers can use these insights for inventory planning and seasonal promotional campaigns.

### **Visualization 6 – Sales by Customer Demographics (Pie Chart)**

A Pie Chart displayed the contribution of different Customer Demographics to overall sales. Each segment represented a customer group and its corresponding sales percentage. This helped identify key customer segments and supported targeted marketing initiatives.

### **Visualization 7 – Foot Traffic Analysis (Bubble Chart)**

A Bubble Chart analyzed the relationship between Foot Traffic and Sales Volume. Bubble size represented sales magnitude while colors differentiated categories. This visualization demonstrated how increased customer traffic influenced product sales and store performance.

### **Visualization 8 – Product Placement Heat Map (Heat Map)**

A Heat Map compared Product Position with Product Category using color intensity to represent Sales Volume. Darker colors indicated stronger performance. This visualization quickly highlighted the best-performing product-position combinations and supported better shelf placement decisions.

## **Dashboard Development**

After creating all eight visualizations, they were combined into a single interactive Tableau Dashboard. Layout containers, titles, legends, and formatting were used to organize the dashboard. Interactive filters allowed users to customize the analysis dynamically. The dashboard provided a comprehensive overview of sales, pricing, promotions, customer behavior, and product placement.

## **Interactive Dashboard Filters**

Filters such as Product Category, Promotion Status, Product Position, Customer Demographics, Seasonal Demand, and Foot Traffic were added to improve interactivity. Selecting any filter automatically updated all related visualizations. These filters enabled users to perform customized analysis based on business requirements.

## **Story Development**

A Tableau Story was created to present the analysis in a sequential and presentation-friendly format. Multiple Story Points explained pricing analysis, sales trends, customer demographics, promotions, seasonal demand, and product placement. Captions were added to describe the purpose of each story point, making the findings easier to understand for stakeholders.

## **Importance of Visualization**

Visualization converts complex numerical data into meaningful graphical representations. Interactive charts help users quickly identify trends, comparisons, and relationships that would otherwise be difficult to observe from tables alone. The combination of eight visualizations, an interactive dashboard, and a Tableau Story provided a complete Business Intelligence solution for retail product placement analysis.

## **Publishing to Tableau Public**

After completing the dashboard and Tableau Story, the workbook was published using Tableau Public. The File > Save to Tableau Public option was selected, and the project was uploaded after signing in with a Tableau Public account. Publishing generated an online workbook that could be shared using a public URL and embed code. This allowed the dashboard and story to be accessed from any web browser while preserving interactivity such as filters and story navigation.

## **Verification of Published Dashboard and Story**

The published workbook was verified by opening the Tableau Public link in a web browser. Each worksheet, dashboard, and story point was checked to ensure it loaded correctly. Interactive filters, legends, and navigation buttons were tested. This verification confirmed that the online version matched the Tableau Desktop version without missing visualizations or errors.

## **Bootstrap Website Development**

A responsive project website was created using HTML5, CSS3, Bootstrap 5, and Visual Studio Code. The website provides an attractive interface for presenting the Tableau Dashboard and Story. Bootstrap components were used to build a navigation bar, hero section, project description, objectives, technology section, embedded dashboard, embedded story, team section, and footer. Responsive design ensured compatibility across desktops, tablets, and mobile devices.

## **Embedding Tableau Dashboard and Story**

The Share option in Tableau Public was used to obtain the embed code for both the dashboard and the story. The generated HTML code was copied and pasted into the Bootstrap webpage. Separate sections were created for the dashboard and story so users could interact with both visualizations without leaving the website.

## Website Testing

The completed website was tested using Google Chrome. Navigation links, embedded Tableau content, and responsive layouts were verified. Dashboard filters and story navigation were checked to ensure proper functionality. The website successfully displayed all project components across different screen sizes.

## Results and Discussion

The dashboard provided a consolidated view of sales performance, pricing strategies, promotions, seasonal demand, and customer demographics. Interactive filters enabled users to explore specific business scenarios. The Tableau Story presented findings sequentially, making the analysis easier to understand. The project demonstrated how Business Intelligence supports informed retail decision-making.

## Key Findings

The analysis showed that product placement significantly influences sales performance. Promotional campaigns positively affected several product categories, while competitor price comparison identified opportunities for pricing improvements. Seasonal demand patterns highlighted the need for effective inventory planning, and customer demographic analysis supported targeted marketing strategies.

## Conclusion

The project successfully demonstrated the complete Business Intelligence lifecycle, from dataset collection and preprocessing to visualization, dashboard creation, story development, publishing, and website integration. Tableau transformed raw retail data into meaningful insights that support better business decisions. The Bootstrap website provided a professional platform for presenting the completed project online.

## Future Scope

Future enhancements may include integrating real-time retail databases, applying machine learning models for sales prediction, deploying dashboards on enterprise cloud platforms, and developing mobile applications for monitoring retail performance. Additional recommendation systems and customer segmentation techniques could further improve business value.

## References

Kaggle (Dataset Source); Tableau Desktop Documentation; Tableau Public Documentation; Bootstrap 5 Documentation; HTML5 and CSS3 Documentation (MDN); Visual Studio Code Documentation; Google Chrome; Google Drive.